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5-F

USE-CASE FLOW SPECIFICATION

Use Case ID : UC-04

Use Case Name : Evaluate Feature Flag

Primary Actor : Client Application / SDK

Description

This use case describes how the Client Application / SDK requests and receives the evaluated state of a feature flag based on user targeting rules, percentage-based rollout configuration, and applicable environment overrides.

Preconditions

1. The requested feature flag exists in the Feature Flag & Dynamic Config Manager.
2. The Client Application / SDK can communicate with the system.
3. The request contains a valid feature flag identifier and the required user targeting information.
4. The relevant feature flag configuration and targeting rules have been defined.

Postconditions

1. The Client Application / SDK receives the final evaluated feature flag state.
2. The feature flag state is determined according to the configured targeting rules, rollout configuration, and applicable environment override.
3. No feature flag configuration is modified during the evaluation process.

Main Success Scenario

1. The Client Application / SDK sends a request to evaluate a feature flag.
2. The system receives the feature flag identifier and user targeting information.
3. The system verifies that the requested feature flag exists.

4. The system identifies the environment associated with the request.
5. The system checks whether an applicable environment-specific configuration override exists.
6. The system evaluates the configured user targeting rules.
7. The system determines whether the user belongs to the configured percentage-based rollout cohort.
8. The system determines the final feature flag state based on the applicable configuration.
9. The system returns the evaluated feature flag state to the Client Application / SDK.
10. The Client Application / SDK receives and applies the returned feature flag state.
11. The use case ends successfully.

Alternate Flow A1 — Feature Flag Not Found

A1.1 At Step 3, the system cannot find the requested feature flag.

A1.2 The system returns an error response indicating that the feature flag does not exist.

A1.3 The Client Application / SDK retains its default or previously available configuration.

A1.4 The use case ends.